

A Similarity Proof of Conjecture 6.3 in *Sharp Schoenberg Type Inequalities and the de Bruin–Sharma Problem*

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Status

full_solution_likely_valid. This packet proves Conjecture 6.3 of Tang–Zhang, *Sharp Schoenberg type inequalities and the de Bruin–Sharma problem*, arXiv:2508.10341v3. The source verifies the conjecture only for $n = 2$; the argument below proves it for every n , and gives the equality characterization.

1 Source Problem

The source considers normalized polynomials

$$P(z) = z \prod_{j=1}^n (z - z_j), \quad z_j \in \mathbb{C} \setminus \{0\},$$

and denotes by $w_1, \dots, w_n$ the zeros of $P'(z)$. Conjecture 6.3 states that

$$\sum_{k=1}^n \frac{1}{|w_k|^2} \leq 3 \sum_{j=1}^n \frac{1}{|z_j|^2} + \left| \sum_{j=1}^n \frac{1}{z_j} \right|^2, \quad (1)$$

and that equality holds whenever the points $z_1, \dots, z_n$ all lie on a single line through the origin.

Conjecture 6.3. *Let*

$$P(z) = z \prod_{j=1}^n (z - z_j), \quad z_j \in \mathbb{C} \setminus \{0\},$$

and let $w_1, \dots, w_n$ denote the zeros of $P'(z)$. Then

$$\sum_{k=1}^n \frac{1}{|w_k|^2} \leq 3 \sum_{j=1}^n \frac{1}{|z_j|^2} + \left| \sum_{j=1}^n \frac{1}{z_j} \right|^2. \quad (6.9)$$

Moreover, equality in (6.9) holds whenever $z_1, \dots, z_n$ all lie on the same line passing through the origin.

In the case $n = 2$ we can verify Conjecture 6.3 by an explicit computation.

2 Idea of the Proof

The source already gives the decisive matrix model. If

$$u_j = \frac{1}{z_j}, \quad u = (u_1, \dots, u_n)^T,$$

then $1/w_1, \dots, 1/w_n$ are the eigenvalues of

$$B = \text{diag}(u) + u\mathbf{1}^T = \text{diag}(u)(I + J),$$

where $J = \mathbf{1}\mathbf{1}^T$. The source estimates these eigenvalues by the Frobenius norm of B , which gives the coarser constant $n + 3$.

The small twist is to keep the same eigenvalues but change the Frobenius norm. Let $C = I + J$ and $H = C^{1/2}$. Since C is positive definite,

$$B = \text{diag}(u)C$$

is similar to

$$A = H \text{diag}(u)H.$$

Schur's inequality bounds the squared moduli of the eigenvalues by $\|A\|_F^2$, and $\|A\|_F^2$ is exactly the right-hand side of (1). Equality is Schur equality, i.e. normality of A , and this normality condition says precisely that all u_j , equivalently all z_j , are collinear with the origin.

3 Proof

We use the following standard form of Schur's inequality.

Lemma 1 (Schur inequality). *Let $A \in M_n(\mathbb{C})$ have eigenvalues $\lambda_1, \dots, \lambda_n$. Then*

$$\sum_{k=1}^n |\lambda_k|^2 \leq \text{tr}(AA^*) = \|A\|_F^2.$$

Equality holds if and only if A is normal.

Theorem 1. *Let*

$$P(z) = z \prod_{j=1}^n (z - z_j), \quad z_j \in \mathbb{C} \setminus \{0\},$$

and let $w_1, \dots, w_n$ be the zeros of $P'(z)$. Then

$$\sum_{k=1}^n \frac{1}{|w_k|^2} \leq 3 \sum_{j=1}^n \frac{1}{|z_j|^2} + \left| \sum_{j=1}^n \frac{1}{z_j} \right|^2.$$

Moreover, equality holds if and only if $z_1, \dots, z_n$ lie on one line through the origin.

Proof. Set $u_j = 1/z_j$, $u = (u_1, \dots, u_n)^T$, $D = \text{diag}(u)$, and $J = \mathbf{1}\mathbf{1}^T$. The matrix model used in the source's proof of Theorem 6.1 shows that the reciprocal critical points $1/w_1, \dots, 1/w_n$ are the eigenvalues of

$$B = D + u\mathbf{1}^T = D(I + J).$$

Put $C = I + J$. The matrix C is positive definite, so $H = C^{1/2}$ is invertible. The matrix

$$A = HDH$$

is similar to $B = DC$, because

$$H^{-1}AH = H^{-1}(HDH)H = DC = B.$$

Thus A and B have the same eigenvalues. By Lemma 1,

$$\sum_{k=1}^n \frac{1}{|w_k|^2} = \sum_{k=1}^n |\lambda_k(A)|^2 \leq \text{tr}(AA^*).$$

We now compute the last trace:

$$\text{tr}(AA^*) = \text{tr}(HDH^2D^*H) = \text{tr}(CDCD^*).$$

Since $C = I + J$,

$$\text{tr}(CDCD^*) = \text{tr}(DD^*) + \text{tr}(JDD^*) + \text{tr}(DJD^*) + \text{tr}(JDJD^*).$$

The first three terms are all $\sum_j |u_j|^2$. For the last term,

$$\text{tr}(JDJD^*) = (\mathbf{1}^T D \mathbf{1})(\mathbf{1}^T D^* \mathbf{1}) = \left| \sum_{j=1}^n u_j \right|^2.$$

Therefore

$$\text{tr}(AA^*) = 3 \sum_{j=1}^n |u_j|^2 + \left| \sum_{j=1}^n u_j \right|^2,$$

which is exactly (1) after substituting $u_j = 1/z_j$.

It remains to discuss equality. By Lemma 1, equality in (1) holds if and only if A is normal. Since H is positive and invertible,

$$AA^* = A^*A \iff DCD^* = D^*CD.$$

The diagonal terms $DD^* = D^*D$ cancel, so this is equivalent to

$$DJD^* = D^*JD.$$

Entrywise, this says

$$u_i \overline{u_j} = \overline{u_i} u_j \quad (1 \leq i, j \leq n).$$

Since no u_j is zero, the last condition is equivalent to every quotient u_i/u_j being real, i.e. to the points $u_1, \dots, u_n$ lying on a single real line through the origin. Taking reciprocals preserves this property, so this is equivalent to $z_1, \dots, z_n$ lying on a single line through the origin. $\square$

4 Verification Notes

The proof uses only:

- the source paper’s reciprocal-critical-point matrix model $B = \text{diag}(u) + u\mathbf{1}^T$;
- similarity by the positive square root of $I + J$;
- Schur’s eigenvalue-singular-value inequality.

The script `code/verify_c1_random.py` performs a randomized numerical check for the equivalent matrix inequality.

5 Novelty Check

Local index searches found no exact packet for arXiv:2508.10341 or for Conjecture 6.3. Web searches on 2026-07-18 for exact phrases around “Conjecture C1”, “Conjecture 6.3”, the reciprocal critical-point inequality, and the matrix forms $\text{diag}(u) + u\mathbf{1}^T$ and $D^{-1} + b\mathbf{1}^T$ found the source arXiv/J-GLOBAL records and adjacent Schoenberg-type material, but no exact solution or this similarity argument. Novelty remains subject to human literature review.

6 References

1. Q. Tang and T. Zhang, *Sharp Schoenberg type inequalities and the de Bruin–Sharma problem*, arXiv:2508.10341v3, 2025.
2. I. Schur, standard eigenvalue majorization inequality; also used in the source paper as the eigenvalue–singular-value comparison lemma.

7 Human Review Recommendation

Review as a likely full solution to Conjecture 6.3. The key points to verify are the source matrix model for reciprocal critical points, the similarity $D(I+J) \sim (I+J)^{1/2}D(I+J)^{1/2}$, and the equality characterization via normality.